

Gitesh Chawda

AI/ML Engineer | Software Developer | Open Source Developer

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SUMMARY

Machine Learning Engineer with industry experience building, securing, and deploying production web applications on AWS. Author and maintainer of open-source ML libraries covering transformer architectures, LLM inference, quantization, and GPU kernel optimization across PyTorch, TensorFlow, and JAX. Active contributor to the Hugging Face and Keras ecosystems with a strong background in deep learning, computer vision, and agentic AI systems.

EXPERIENCE

ZeroModels, Creater and Maintainer, Open Source Project (Aug 2024 - Present) [Github](#)

- Created a backend-agnostic, pip-installable Keras 3 library spanning 100+ vision, speech, LLM, and VLM architectures (SAM, DETR, DINOv3, Whisper, Llama, Qwen, DeepSeek, Mistral, Gemma, GLM) with single-codebase execution on TensorFlow, PyTorch, and JAX.
- Engineered Hugging Face-to-Keras checkpoint conversion with bit-close parity validation against reference implementations.
- Introduced Keras 3 native weight-only quantization (INT8, INT4, FP8) using pure keras.ops, enabling up to 8x model size reduction for efficient inference deployment.
- Grew adoption to 7k+ monthly PyPI downloads with automated testing, releases, and CI/CD via GitHub Actions.

Open Source Contributor, Hugging Face Transformers, Keras, Keras.io, KerasCV (Apr 2022 - Present)

- Authored 42 merged PRs across Hugging Face Transformers, Keras, Keras.io, and KerasCV, improving model APIs, tokenizer design, type safety, and developer workflows.
- Added 10+ Keras 3 multi-backend APIs, layers, and utilities including PSNR, unravel_index, diagflat, RandomGrayscale, Equalization, and image augmentation workflows.
- Contributed ConvMixer models, YOLOv8 improvements, CIoU loss, detection tests, and official keras.io examples.

Fiserv, Software Developer, Chennai, India (Jul 2023 - Aug 2024)

- Developed and maintained a production web application with React.js, GraphQL, and AWS (AppSync, Lambda, API Gateway, Cognito, S3), powering internal banking and payments workflows.
- Strengthened application security with client-side data tampering prevention and request validation, and partnered with AppSec teams to remediate vulnerabilities found in audits and penetration testing.
- Managed 20+ production deployments and critical hotfixes across development, staging, and production environments while ensuring reliable application availability.

Braynix AI, Deep Learning Intern, Bangalore, India (Sep 2021 - Dec 2021)

- Built deep learning pipelines for medical X-ray abnormality detection, raising validation accuracy by 20% over baseline CNN models on 5,000+ images.
- Reached 92% precision and 0.85+ IoU in medical image segmentation using a U-Net with an EfficientNet-B4 encoder.
- Trained and benchmarked 3 object detection models (YOLOv5, SSD, RetinaNet) on 1,000+ labeled images at 0.78+ mAP.

PROJECTS

Trinix: Triton-accelerated PyTorch optimization library with custom GPU kernels, including fused attention variants (MHA, GQA, MQA), RMSNorm, LayerNorm, SwiGLU, RoPE, and ALiBi, delivering up to 4.8x faster execution. [Github](#)

InferML: Local-first inference server for 143+ Hugging Face model families (LLMs, VLMs, diffusion, ASR, TTS) with an OpenAI-compatible API, built on FastAPI and React. [Github](#)

OpenRAG: Local-first RAG framework with switchable vector (Chroma) and knowledge graph (Neo4j) retrieval, cross-encoder reranking, and local LLMs. [Github](#)

txtcode: TypeScript framework for orchestrating AI coding agents across 6 messaging platforms and 10 LLM backends with context-preserving switching. [Github](#)

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

ML and Deep Learning: PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, scikit-learn, Computer Vision, Diffusion Models

LLM Training and Inference: vLLM, TensorRT-LLM, Triton, CUDA, XLA, Quantization, LoRA, PEFT, SFT, GRPO, KV Cache Optimization, FlashAttention, bitsandbytes, Model Serving

Agentic AI: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, OpenAI Agents SDK, MCP, A2A

MLOps and Cloud: Docker, Linux, Git, GitHub Actions, CI/CD, AWS, GCP

Web and Data: React.js, Flask, FastAPI, GraphQL, Node.js, NumPy, Pandas, OpenCV, MySQL, MongoDB, Oracle

Coding Agents: Claude Code, Codex, OpenCode

EDUCATION

California State University, Fullerton, Master of Science in Computer Science, California, USA (Aug 2024 - May 2026)

Pimpri Chinchwad College of Engineering, Bachelor of Engineering in Computer Science, Pune, India (Aug 2019 - Jul 2023)

PUBLICATIONS

EffiConvRes: An Efficient Convolutional Neural Network with Residual Connections and Depthwise

Convolutions, 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBE). Paper

Survey and Comparison of Various Pre-trained CNN Architectures and CNN-Transformer Combinations, Smart Grid and Electric Vehicle Conference. [Paper](#)